
DistillRouter: Efficient LLM Routing via Knowledge Distillation

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 LLM routing improves the cost-performance tradeoff of large-scale LLM serving
2 by directing queries to the least expensive capable model. However, routing to
3 an appropriate model with balanced inference cost trade become deployment bot-
4 tlenecks. We propose DistillRouter, a routing-policy distillation framework that
5 transfers routing policy from an expensive teacher router to a lightweight distilled
6 router. DistillRouter converts routing signals into a supervised classification prob-
7 lem, enabling deployment through a single forward pass while preserving routing
8 quality. We evaluate DistillRouter on GSM8K and MATH under both binary and
9 three-label routing settings using multiple distilled-router backbone scales. Across
10 all configurations, distilled routers achieve routing accuracy comparable to or better
11 than the teacher while reducing routing latency by up to 153×. We further investi-
12 gate reasoning-augmented supervision and find that although reasoning traces are
13 associated with improved policy transfer at larger model scales, this improvement is
14 not consistently associated with better routing correctness. Our results indicate that
15 routing policies can be effectively compressed into compact models, suggesting a
16 practical approach for reducing routing overhead in large-scale LLM serving.

17 1 Introduction

18 Large Language Models (LLMs) now underpin a wide range of applications, from conversational
19 assistants and coding tools to search and enterprise copilots. Because inference cost and latency scale
20 with model size, production systems serving large query volumes face mounting pressure to control
21 both under tight responsiveness constraints.

22 LLM workloads are highly heterogeneous in difficulty: some queries demand a frontier-scale model’s
23 reasoning, while many others are answered just as well by a much smaller one. Sending every query
24 to the most capable model therefore wastes compute and inflates serving cost. LLM routing addresses
25 this by dynamically directing each query to the cheapest model expected to answer it correctly,
26 substantially reducing cost while preserving response quality.

27 Routing itself, however, introduces a new inference problem: deciding which model should answer a
28 query. Existing approaches learn this policy from preference data, reward signals, or reinforcement
29 learning, and the router that makes this decision varies widely in cost — from a lightweight classifier
30 to a full LLM that reasons over each routing decision at inference time [Ong et al., 2024, Zhang et al.,
31 2025] (Appendix A.1 surveys these approaches in detail). Such approaches can improve routing
32 accuracy, but when the router is itself a large, expensive model, it reintroduces the very inference
33 cost routing is meant to avoid — the router’s own decision-making now adds latency and compute to
34 every incoming query.

35 This raises a natural question:

Can the routing policy learned by an expensive model be compressed into a lightweight model?

Knowledge distillation [Hinton et al., 2015] provides a direct mechanism for this: a teacher model’s behavior is used to supervise a smaller student model, which learns to approximate the teacher’s function without inheriting its computational cost. Applied to routing, an expensive teacher router is queried offline over a large set of training queries to produce routing decisions; these decisions serve as training labels for a compact distilled router, which learns to reproduce the teacher’s routing policy and performs routing through a single forward pass at deployment time. This confines the teacher’s cost to an offline labeling step, so only the lightweight distilled router is queried at inference.

We introduce DistillRouter, a framework that transfers a routing policy from an expensive teacher to a compact distilled router. We study two distillation strategies: route-only distillation, which transfers the teacher’s routing decisions directly, and reasoning-plus-route distillation, which additionally supervises the router with teacher-generated reasoning traces as an auxiliary training signal, at no extra deployment cost. Comparing the two isolates whether richer supervision improves distillation.

Evaluated on two benchmarks, DistillRouter compresses routing policies into models that reduces inference latency while matching the teacher’s routing accuracy. We study the role of reasoning supervision throughout, evaluating both how closely a distilled router imitates its teacher and how well it performs against independent oracle labels. Our contributions are:

- **DistillRouter:** a knowledge-distillation framework for compressing expensive LLM routing policies into lightweight, deployable routers.
- **Efficient Routing via Distillation:** distilled routers transfer routing policy effectively, cutting overhead while matching or exceeding the teacher’s routing accuracy.
- **Understanding Reasoning Supervision:** an empirical study showing that reasoning supervision improves policy transfer only at larger model scales.

2 Problem Formulation

Consider a routing environment consisting of a set of candidate language models

$$\mathcal{M} = \{m_1, m_2, m_3\},$$

ordered by increasing parameter count, capability, and serving cost. Given an incoming user query x , the goal of the routing system is to predict an appropriate routing decision

$$y \in \mathcal{Y},$$

that determines which model category should process the query. The routing objective is to assign each query to the least expensive model category capable of producing a correct answer. Routing decisions therefore reflect the capability level required to answer a query correctly: queries that can be handled by lower-cost models should be assigned lower-cost routing decisions, while more challenging queries should be assigned higher-capability routing decisions.

An expensive *teacher router* defines a routing policy

$$\pi_T(y \mid x),$$

which maps queries to routing decisions. The teacher may employ larger language models, multi-stage prompting, or explicit reasoning to determine the appropriate routing decision. Our objective is to learn a lightweight *distilled router*

$$\pi_\theta(y \mid x),$$

that approximates the teacher’s routing policy while operating at substantially lower inference cost. Formally, the problem addressed in this work is:

Given an expensive routing policy π_T , can we learn a lightweight routing policy π_θ that preserves routing accuracy while significantly reducing routing latency and computational overhead?

DISTILLROUTER is independent of the specific routing label space. In this work, we evaluate two routing settings.

80 2.1 Binary Routing

81 Our primary configuration uses a binary label space

$$\mathcal{Y}_{\text{binary}} = \text{small}, \text{large}.$$

82 Here, the `small` label represents queries whose cheapest correct assignment belongs to either the
83 m_1 or m_2 category, while `large` represents queries that require the m_3 category. This formulation
84 focuses on the key routing distinction between queries that can be handled by lower-cost models and
85 those requiring the highest-capability model.

86 2.2 Three-Label Routing

87 To evaluate finer-grained routing decisions, we additionally consider

$$\mathcal{Y}_{\text{3label}} = \text{small}, \text{medium}, \text{large}.$$

88 In this setting, the router directly predicts the model assigned to a query, enabling more precise model
89 selection across the candidate pool. The binary and three-label settings evaluate different levels of
90 routing granularity: binary routing measures whether the router can distinguish between lower-cost
91 and highest-capability model categories, whereas three-label routing evaluates whether the router can
92 accurately differentiate among all available model categories.

93 2.3 Oracle Labels

94 For each query, the oracle label is defined as the smallest model category that produces the correct
95 answer under our evaluation protocol. Specifically, given the candidate model set $\{m_1, m_2, m_3\}$, we
96 execute all candidate models and assign the query to the least expensive model whose output matches
97 the reference answer. If only the largest model answers correctly, the oracle label is `large`; if both
98 m_1 and m_2 answer correctly, the oracle label is `small`. These oracle labels represent the optimal
99 cost-aware routing decision for each query. Oracle labels are used exclusively for evaluation and
100 provide an independent measure of routing accuracy — they are never used for training.

101 3 Methodology

102 3.1 DistillRouter Framework

103 DistillRouter transfers a routing policy from an expensive teacher router to a lightweight distilled
104 router. Given a query x , the teacher estimates the least expensive model category required to answer
105 it and generates supervision for training; the distilled router then learns this policy and replaces the
106 teacher at deployment time. The pipeline has two stages: **teacher-policy generation**, where the
107 teacher processes training queries and produces routing supervision, optionally including reasoning
108 traces that explain its decision; and **policy transfer**, where the distilled router is trained to predict the
109 teacher’s labels. Only the distilled router is deployed, performing routing in a single forward pass:
110 latency and compute drop substantially, while routing accuracy is preserved or improved relative to
111 the teacher.

112 3.2 Routing-Policy Distillation

113 Let $x \in \mathcal{X}$ be a query and $\mathcal{Y}$ the routing label space defined in Section 2 (either $\mathcal{Y}_{\text{binary}}$ or $\mathcal{Y}_{\text{3label}}$).
114 Given a teacher policy $\pi_T(y \mid x)$, DistillRouter learns $\pi_\theta(y \mid x) \approx \pi_T(y \mid x)$ from teacher-generated
115 supervision; once trained, the distilled router routes without consulting the teacher.

116 3.3 Teacher Router

117 The teacher supplies routing supervision in two sequential steps. First, *reasoning generation*: it
118 analyzes the query and produces a reasoning trace $r = f_{\text{reason}}(x)$ describing the query’s characteristics;
119 these traces supervise training only and are never used at deployment. Second, *routing prediction*:
120 conditioned on the query, the trace, and few shot examples, it predicts a routing label $y = f_{\text{route}}(x, r)$,
121 the model category expected to suffice. Applying this across the training corpus yields $D_T =$
122 $\{(x_i, y_i, r_i)\}_{i=1}^N$, the supervision used for distillation.

123 3.4 Distilled Router

124 The distilled router is initialized from a pretrained language-model backbone. For a query $x =$
 125 $(t_1, \dots, t_n)$, the decoder-only backbone produces hidden states $H = (h_1, \dots, h_n)$; following prior
 126 work adapting decoder-based language models to non-generative tasks [Suganthan et al., 2025], we
 127 take the final-token representation $h_{pool} = h_n$ and pass it through a lightweight classification head:

$$h' = \text{Dropout}\left(\text{GELU}(W_1 \text{LayerNorm}(h_{pool}) + b_1)\right), \quad \hat{y} = \text{Softmax}(W_2 h' + b_2), \quad (1)$$

128 with learnable W_1, W_2, b_1, b_2 , LayerNorm [Ba et al., 2016], GELU [Hendrycks and Gimpel, 2016],
 129 and Dropout [Srivastava et al., 2014]. The resulting $\hat{y} = \pi_\theta(y | x)$ is the distilled router’s prediction.

130 3.5 Route-Only Distillation

131 The first strategy transfers only the teacher’s routing decisions, via

$$L_{\text{route}} = - \sum_{c \in \mathcal{Y}} y_c \log \hat{y}_c, \quad (2)$$

132 where y_c is the teacher label and $\hat{y}_c$ the predicted probability. This is the primary distillation baseline.

133 3.6 Reasoning-Plus-Route Distillation

134 A routing label supervises only the teacher’s final decision, so we test whether adding the teacher’s
 135 reasoning traces as supervision helps. The distilled router is additionally trained to predict the
 136 reasoning sequence $r = (r_1, \dots, r_T)$:

$$L_{\text{reason}} = - \sum_{t=1}^T \log P(r_t | r_{<t}, x). \quad (3)$$

137 The two terms combine as

$$L = \lambda_{\text{route}} L_{\text{route}} + \lambda_{\text{reason}} L_{\text{reason}}. \quad (4)$$

138 Since reasoning traces are used only in training, both variants have identical deployment cost: at
 139 inference, the router sees only the query and predicts a label in one pass.

140 4 Experimental Setup

141 The goal of our experiments is to evaluate whether routing policies can be effectively compressed
 142 through routing-policy distillation and whether reasoning supervision improves policy transfer (4.1).

143 4.1 Research Questions

144 We design our experiments to answer three questions.

145 **RQ1.** How effectively can a routing policy be transferred from an accurate but expensive teacher
 146 router to a lightweight distilled router?

147 **RQ2.** Does augmenting routing supervision with teacher reasoning improve policy transfer relative
 148 to route-only distillation?

149 **RQ3.** Do improvements in policy transfer translate into improvements in routing accuracy under
 150 independent oracle labels?

151 Answering RQ3 requires an evaluation target that does not simply re-measure performance against
 152 the teacher router. This is the role of the oracle labels (2.3). Oracle labels are used only for evaluation
 153 and provide an independent measure of routing accuracy; they are never used to supervise a router
 154 during training.

155 4.2 Routing Label Spaces

156 DistillRouter (§3) is agnostic to the routing label space $\mathcal{Y}$. This section fixes the two routing settings
 157 evaluated in this paper as experimental configurations rather than methodological choices.

158 **Binary Routing.** Our primary configuration frames routing as an escalation decision,

$$\mathcal{Y}_{\text{binary}} = \{\text{small}, \text{large}\}, \quad (5)$$

159 where `small` denotes routing to whichever of $\{m_1, m_2\}$ is the cheapest candidate that answers
 160 the query correctly, and `large` denotes routing to m_3 . Oracle labels (2.3) determine the correct
 161 assignment. This routing setting reflects the deployment scenario most relevant to cost-sensitive
 162 routing: whether a query can remain on a low-cost model or must be escalated to the highest-capability
 163 model.

164 **Three-Label Routing.** To evaluate finer-grained routing decisions, we additionally consider a
 165 three-label routing label space that maps directly to the three candidate tiers in $\mathcal{M}$. This setting is
 166 evaluated as a generalization check rather than the primary experimental configuration.

167 **Model-Scale Analysis.** Independently of routing setting, we repeat experiments using both 270M
 168 and 1B distilled-router backbones to assess whether routing-policy distillation behaves differently
 169 across model scales (A.8). This analysis is performed under both routing settings, completing the full
 170 routing-setting $\times$ backbone grid.

171 4.3 Routing Environment

172 A routing environment fixes three components before introducing either router: the queries a router
 173 must classify, the candidate models available to answer them, and the routing label space used to
 174 represent routing decisions (2).

175 **Datasets.** We evaluate DistillRouter on two mathematical question-answering benchmarks:
 176 GSM8K [Cobbe et al., 2021], consisting of grade-school arithmetic word problems, and MATH
 177 [Hendrycks et al., 2021], consisting of competition-level mathematics problems spanning seven
 178 subjects and five difficulty levels.

179 Split sizes and per-tier oracle-label distributions are in Appendix Table 4; GSM8K is dominated
 180 by the medium tier, whereas MATH is predominantly assigned to the large tier, reflecting greater
 181 routing difficulty in the latter dataset. Reference answers were extracted and normalized prior to
 182 evaluation: GSM8K answers were parsed from the `<answer>` delimiter, whereas MATH answers
 183 were extracted from `\boxed{\}` expressions. Exact-match evaluation was used throughout, without
 184 symbolic-equivalence checking.

185 **Candidate Models.** Three computational tiers (small/medium/large) plus the teacher span two open-
 186 weight families, Gemma 3 [Gemma Team, 2025] and Qwen2.5 [Qwen Team, 2024], chosen to reflect
 187 heterogeneous deployment environments and remain efficient enough for exhaustive oracle-label
 188 construction (2.3); full model names, parameter counts, and roles are in Appendix Table 5.

189 **Why a 3B, non-frontier teacher.** We use Qwen2.5-3B-Instruct rather than a frontier API model
 190 (GPT-4o, Gemini, Claude) to keep the pipeline self-hosted and reproducible, since our question
 191 is whether an existing router’s policy can be distilled, not what the most accurate router is. The
 192 two-stage prompt (A.4) was fixed once, not tuned for accuracy, so the 0.592 oracle accuracy in
 193 Section 5.1 is a non-optimized baseline, not a ceiling; whether findings hold with a frontier-scale
 194 teacher is open (6).

195 **Routing Setting.** Binary routing is used throughout Sections 5.1–5.2 as the primary configuration.
 196 Three-label routing is evaluated separately as a generalization check. Both settings are defined in
 197 §4.2.

198 Teacher prompting details and distilled-router training hyperparameters are in Appendix A.4 and A.5.

199 4.4 Evaluation Metrics

200 We evaluate distilled routers along three dimensions: *policy transfer* (accuracy and macro-F1 against
 201 the teacher’s routing decisions, answering RQ1–RQ2), *routing accuracy* (accuracy and macro-F1
 202 against independent oracle labels, answering RQ3), and *deployment efficiency* (routing latency and
 203 speedup relative to the teacher, excluding tokenization, model loading, and answer-generation latency).
 204 Full metric definitions, including how routing latency is measured, are in Appendix A.6.

5 Results

This paper’s central claim is that policy distillation enables efficient LLM routing: a compact router using one forward pass can match or exceed an expensive multi-stage teacher’s routing accuracy at a fraction of the deployment cost. We evaluate routing accuracy (RQ1, Section 5.1), then deployment efficiency (Section 5.2), then the effect of reasoning supervision on policy transfer (RQ2, Section 5.3). We close by examining why distilled routers exceed the teacher’s oracle accuracy (Section 5.4) and whether policy transfer predicts routing accuracy (RQ3, Section 5.5), with scaling-behavior and error-analysis checks in Appendix A.8–A.9. Unless noted, results use the primary configuration: binary routing, 270M backbone.

5.1 Routing Accuracy

RQ1: Can routing policies be effectively transferred from a teacher router to a distilled router?

We report routing accuracy, downstream answer quality, and latency for the teacher and both distilled routers in Table 1, under the primary binary-routing configuration. Distillation substantially improves routing accuracy over the teacher, and does so while cutting routing latency by roughly two orders of magnitude and using a far smaller deployed model: the distilled variants dominate the teacher on every axis measured. Several additional observations follow.

Table 1. Routing Accuracy and Deployment Efficiency (binary label space, 270M backbone; primary configuration). Accuracy is measured against oracle labels; end-to-end accuracy resolves each decision to a candidate and checks answer correctness (4.4). Pooled across both datasets (1,000 queries/seed), mean $\pm$ SD across 10 seeds. Always-Medium/Large serve every query with one fixed candidate (0 ms latency); their Acc vs. Oracle is that constant-label predictor’s accuracy against oracle ground truth.

Method	Acc vs. Oracle	End-to-End Acc	Routing Latency (ms)
Always Medium	0.349	0.336	–
Always Large	0.651	0.456	–
Teacher	0.592	0.392	1,633
Route-only	0.682 ± 0.011	0.405 ± 0.005	10.9
Reasoning-plus-route	0.692 ± 0.008	0.408 ± 0.005	11.2

First, the gain is not just in oracle accuracy: the corresponding improvement in end-to-end accuracy is real, though more modest, since downstream quality also depends on the candidate models themselves, not just the routing decision. This still indicates that better routing decisions bring practical benefits beyond routing metrics alone. The teacher’s own accuracy is a deliberately non-optimized baseline (4.3), so the relevant comparison throughout is the gap between the teacher’s policy and what distillation does to it, not the best achievable absolute number.

Second, this advantage is not confined to the primary configuration. Figure 1 extends the comparison to all eight evaluated distilled-router settings: every one achieves both lower latency and higher oracle F1 than the teacher anchor for its own label space, placing all eight on a strictly superior Pareto frontier.

Third, routing latency alone does not capture the full cost of serving a query: the routing decision’s real purpose is to control the cost of *answering* it, which also depends on the answer-generation latency of whichever candidate the decision selects. Figure 2 places DistillRouter and the teacher against three fixed-policy baselines on this fuller cost (routing latency plus the expected answer-generation latency of the served candidate, weighted by the predicted-label distribution) versus end-to-end accuracy.

On this view, the advantage narrows: both the teacher and DistillRouter fall between the Always-Medium and Always-Large baselines rather than beating either, and Always-Large retains the single highest raw accuracy of any point plotted. Routing therefore does not dominate the static Always-Large baseline in this aggregate view. Its value lies instead in the oracle-accuracy results above: routing identifies *which* queries can safely stay on a cheaper candidate, a distinction the three-point

Figure 1. F1-Latency Frontier. Routing Accuracy (y-axis) vs. per-query routing latency in ms, log scale (x-axis), for all eight distilled-router configurations plus the teacher, each in its own label space. Color: distillation variant. Shape: backbone size (270M circle, 1B square). Fill: label space (binary filled, three-label hollow). Error bars are ± 1 SD across 10 seeds. Dashed/dotted lines mark the teacher’s own oracle macro-F1 in binary (0.591, distinct from but close to the 0.592 accuracy figure in Table 1) and three-label (0.380); since these are not on a comparable scale, each family is judged only against its own teacher anchor. Every distilled-router point beats its label-space’s teacher on both latency and F1: all eight lie on a superior Pareto frontier.

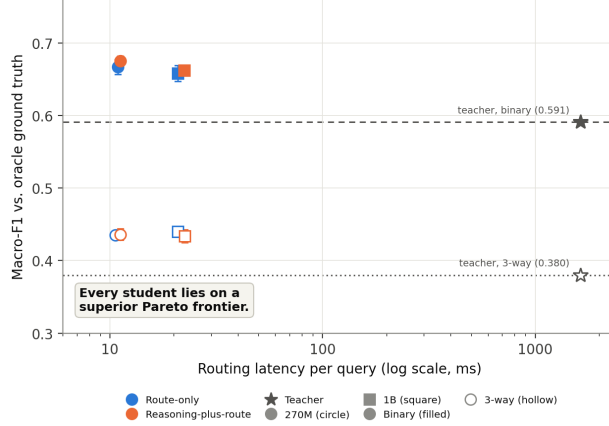
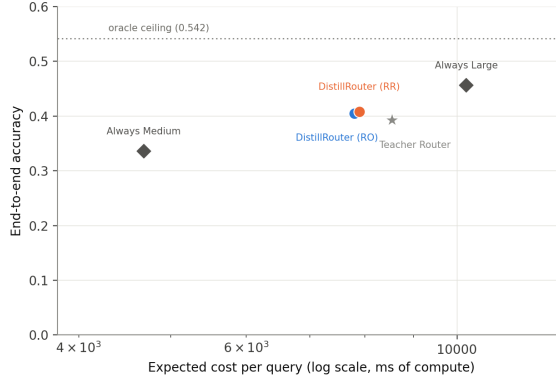


Figure 2. Cost-Quality Frontier (binary label space, 270M backbone). Expected cost per query in milliseconds of compute, log scale (x-axis) vs. end-to-end accuracy (y-axis). Diamonds: fixed policies that serve every query with one candidate. Star: the teacher. Circles: DistillRouter, route-only (RO) and reasoning-plus-route (RR). Dotted line: the oracle ceiling (0.542), the accuracy of always picking the cheapest candidate that happens to answer correctly, an upper bound no real policy can exceed. Cost is measured in compute latency rather than dollar price (4.3).



242 plot cannot show. The gap to the oracle ceiling likewise shows substantial headroom remains for
 243 every policy plotted, including the teacher.

244 Together, these results show that routing policy transfers effectively into lightweight distilled models,
 245 delivering large, consistent oracle-accuracy and latency gains across every configuration tested —
 246 though end-to-end accuracy gains are smaller, and routing does not yet dominate the simplest static
 247 baseline.

248 5.2 Deployment Efficiency

249 How much deployment efficiency can be gained through distillation?

Deployment efficiency is DistillRouter’s primary motivation, and the gains are large: shifting from the teacher’s multi-stage prompting process (3.1) to a single forward pass delivers speedups of up to 153× over the teacher, with routing accuracy at or above the teacher’s throughout (5.1).

This holds across every evaluated configuration (Table 1 for the primary configuration; Appendix Table 6 for the remaining six): label-space granularity has little effect on latency, and although scaling the backbone from 270M to 1B costs more, substantial efficiency gains over the teacher remain.

5.3 Reasoning Supervision and Policy Transfer

RQ2: Does reasoning supervision improve policy transfer?

Having established that distillation works and is efficient, we test whether reasoning supervision improves it further. This section covers policy transfer alone (agreement with the teacher’s labels); Section 5.5 covers whether the decisions are actually correct. Two patterns emerge from Table 2.

Table 2. Policy Transfer Results. Macro-F1 against the teacher’s decision, mean $\pm$ SD across 10 seeds. Δ is route-only minus reasoning-plus-route on macro-F1, with a 95% CI from a paired bootstrap [Koehn, 2004] over the 10 seeds; **bold** marks a CI that excludes zero.

Configuration	RO F1	RR F1	Δ F1 (RO – RR)
Binary (270M)	0.651 $\pm$ 0.005	0.645 $\pm$ 0.010	+0.0065 ([−0.0118, +0.0232])
Three-Label	0.477 $\pm$ 0.025	0.460 $\pm$ 0.018	+0.017 ([−0.022, +0.062])
Binary (1B)	0.652 $\pm$ 0.016	0.676 $\pm$ 0.008	−0.024 ([−0.062, −0.003])
Three-Label	0.505 $\pm$ 0.012	0.529 $\pm$ 0.012	−0.024 ([−0.055, −0.001])

First, scale matters: at the 270M backbone, route-only and reasoning-plus-route distillation transfer the teacher’s policy about equally well, with confidence intervals overlapping on every metric and including zero. At the larger 1B scale, however, reasoning supervision produces a small but statistically reliable gain in policy transfer, consistent across both label spaces. Because this comparison holds data, architecture, and seeds fixed and varies only the training objective, the association rests on firmer footing than a purely observational correlation would, though it still comes from a single teacher and task domain (6). This is consistent with reasoning signals helping the router reproduce teacher behavior more closely as capacity grows.

Second, and more surprisingly, this improved imitation does not translate into better routing accuracy. At 1B, reasoning-plus-route distillation’s end-to-end accuracy is actually slightly *lower* than route-only’s at both label spaces (5.1) — the opposite direction from its confirmed fidelity advantage. A single-configuration study would have reported just one of these 270M/1B outcomes as the general finding.

5.4 Distilled Routers Achieve Higher Oracle Accuracy Than the Teacher

A counterintuitive observation is that distilled routers achieve higher oracle accuracy than the teacher used to generate their supervision, under our evaluation protocol (Table 1, 5.1). Because both are measured against the same ground truth, this is a direct, apples-to-apples comparison, not an artifact of the teacher’s and oracle’s label distributions differing. The distilled router is trained on nothing but the teacher’s own labels, so we do not treat "the student exceeds the teacher" as a settled finding — we report it as a robust empirical pattern, and return to candidate explanations at the end of this section.

First, this pattern generalizes: Table 3 shows the oracle-accuracy point estimate at or above teacher accuracy in all eight configuration/variant pairs, not just the primary one — though not uniformly reliably, since one cell (Binary (1B), reasoning-plus-route) is close enough to call a tie, and is metric-dependent besides: under macro-F1 (Table 3) that same cell actually favors the teacher, consistent with Section 5.5’s broader finding that accuracy and macro-F1 do not always agree. This pattern alone is suggestive but not conclusive on its own, since the teacher and oracle have different label distributions and majority-class baselines; the controlled comparison above (teacher’s own oracle accuracy vs. the distilled routers’, same reference for both) is what actually establishes the claim.

Table 3. Policy Transfer vs. Routing Accuracy, mean $\pm$ SD across 10 seeds, all four configurations, both metrics. In every row, the oracle accuracy point estimate is at least as high as accuracy against the teacher, though for Binary (1B) reasoning-plus-route the two are within 0.002 of each other, well inside one SD (see prose below); **bold** marks the higher-scoring variant in each F1 column, within each configuration, illustrating that accuracy and macro-F1 do not always agree (e.g. Binary (1B) favors reasoning-plus-route on accuracy but route-only on F1).

Configuration	Variant	Acc vs. Teacher	Acc vs. Oracle	F1 vs. Teacher	F1 vs. Oracle
Binary (270M)	Route-only	0.652 $\pm$ 0.005	0.682 $\pm$ 0.011	0.651 $\pm$ 0.005	0.667 $\pm$ 0.010
	Reasoning-plus-route	0.645 $\pm$ 0.011	0.692 $\pm$ 0.008	0.645 $\pm$ 0.010	0.675 $\pm$ 0.007
Three-Label	Route-only	0.549 $\pm$ 0.010	0.636 $\pm$ 0.016	0.477 $\pm$ 0.025	0.435 $\pm$ 0.005
	Reasoning-plus-route	0.545 $\pm$ 0.007	0.645 $\pm$ 0.015	0.460 $\pm$ 0.018	0.436 $\pm$ 0.008
Binary (1B)	Route-only	0.652 $\pm$ 0.016	0.676 $\pm$ 0.014	0.652 $\pm$ 0.016	0.658 $\pm$ 0.011
	Reasoning-plus-route	0.677 $\pm$ 0.008	0.679 $\pm$ 0.011	0.676 $\pm$ 0.008	0.662 $\pm$ 0.007
Three-Label	Route-only	0.574 $\pm$ 0.007	0.633 $\pm$ 0.010	0.505 $\pm$ 0.012	0.440 $\pm$ 0.007
	Reasoning-plus-route	0.587 $\pm$ 0.014	0.629 $\pm$ 0.009	0.529 $\pm$ 0.012	0.434 $\pm$ 0.009

Second, the effect is not simple memorization. The teacher systematically under-predicts queries that require the large-tier candidate, and distilled routers substantially recover these cases (Appendix A.9 gives the confusion-matrix breakdown) — learning a different, more accurate escalation boundary than the one the teacher itself demonstrated, rather than faithfully reproducing its error pattern.

Why would a classifier trained only on the teacher’s own labels outperform that teacher? The mechanism is not established; we see three non-exclusive candidates: (1) *regularization* — fitting one boundary over many examples may average out per-query errors rather than reproduce them; (2) *prompting instability* — the gap may reflect noise in the teacher’s own two-stage prompting (A.4) rather than a general property of distillation; (3) *subtler contamination* — though splits are disjoint and oracle labels are never used for training (2.3), template overlap between GSM8K and MATH cannot be fully ruled out. Distinguishing these, e.g. by retraining against a deliberately degraded teacher, is left to future work.

5.5 Policy Transfer Does Not Predict Routing Accuracy

RQ3: Do improvements in policy transfer translate into improvements in routing accuracy?

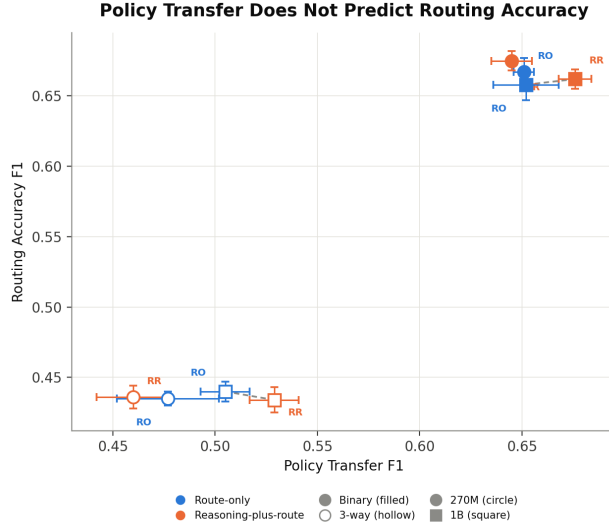
We compare each variant’s policy transfer against its accuracy on oracle labels, and find that imitating the teacher more closely does not reliably mean being more correct.

Across configurations, the variant with higher teacher F1 does not necessarily achieve higher oracle F1: at three of the four configurations, the higher-teacher-F1 variant has *lower* oracle-F1; at the fourth (binary/1B) the rankings agree, but the correctness gap is still not statistically reliable. Reasoning-augmented distillation is associated with higher fidelity at the 1B scale (5.3) but not consistently with higher correctness: no configuration shows a statistically reliable oracle-F1 advantage for either variant, even at the two 1B configurations where the fidelity advantage is confirmed.

To rule out a spurious correlation rather than relying on the eight-point scatter alone, we test this on statistical footing: a naive pooled correlation across all eight points is strongly positive ($r = +0.97$, $p < 0.001$), but this is a structural confound — binary-label-space configurations score higher than three-label ones on *both* axes simply because binary is the easier problem (Table 3) — not evidence that fidelity predicts correctness. Controlling for label space (a partial correlation over all 80 seed runs) and testing the reasoning-vs.-route-only *change* directly (a paired test, one point per configuration) both collapse this to a null relationship, confirming the null correlation in Figure 3 is not an artifact of a sparse scatter; full test statistics are in Appendix A.10.

Policy transfer and routing accuracy capture different aspects of routing behavior; policy transfer alone should not be used as a proxy for routing accuracy. Two further checks support this picture without changing it. First, scaling the distilled-router backbone from 270M to 1B produces no reliable oracle-accuracy advantage in either label space, across the full 2×2 label-space-by-backbone grid — a low-capacity prediction problem for which lightweight models already suffice (Appendix A.8).

Figure 3. Policy Transfer Does Not Predict Routing Accuracy. Teacher F1 (x-axis, fidelity) vs. oracle F1 (y-axis, correctness) for route-only (RO) and reasoning-plus-route (RR) at all four configurations (same marker convention as Figure 1), the full rows of Table 3. Dashed connectors join same-configuration pairs; error bars are ± 1 SD across 10 seeds.



Second, a confusion-matrix analysis pooling all 10 seeds shows the accuracy gains reported above are concentrated in one specific failure mode: the teacher under-escalates large-tier queries at essentially chance level, and distillation corrects this specifically rather than improving uniformly across all decisions (Appendix A.9).

6 Limitations

This study has several limitations. First, experiments are restricted to mathematical reasoning benchmarks, and the extent to which routing-policy distillation generalizes to other domains such as coding, instruction following, or open-domain question answering remains unverified. Second, DistillRouter is evaluated using a single teacher router, a mid-size, self-hosted, non-frontier model (Qwen2.5-3B-Instruct) with a fixed two-stage prompt that was not iterated on to maximize teacher accuracy (4.3); we made this choice deliberately, to keep the pipeline reproducible without paid APIs and to study distillation of a realistic, imperfect router rather than an oracle-level one, but it also means we have not tested whether a substantially stronger or frontier-scale teacher (e.g., GPT-4o, Gemini, or Claude), or a more heavily prompt-engineered version of the same teacher, would change the transfer characteristics we report, including the direction of the student-exceeds-teacher effect in Section 5.4. Third, our scaling analysis includes only two distilled-router backbone sizes, limiting conclusions about how routing-policy distillation behaves across broader capacity ranges. Finally, reasoning is studied exclusively as a training-time supervision signal, and we do not investigate its interaction with inference-time reasoning-based routing methods. These limitations provide several directions for future work.

7 Conclusion

We presented DistillRouter, a routing-policy distillation framework for efficient LLM routing. DistillRouter transfers routing policy from an expensive teacher router into a lightweight distilled model, enabling routing decisions through a single forward pass. Across multiple routing configurations, distilled routers achieved higher oracle-label accuracy than the teacher used to supervise them while reducing routing latency by up to 153 $\times$, though the mechanism behind this gap remains an open question (Section 5.4). We further found that reasoning supervision is associated with improved policy transfer at larger scales but is not consistently associated with improved routing correctness.

Overall, our results support the conclusion that routing policies can be effectively compressed through knowledge distillation, suggesting a practical approach for reducing routing overhead in large-scale LLM deployment.

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 407

408 A Appendix

409 Section A.1 surveys prior work in full. Sections A.2–A.9 extend the main body’s Experimental
 410 Setup and Results sections (dataset splits, candidate models, teacher prompting details, training
 411 hyperparameters, metric definitions, per-configuration efficiency, scaling behavior, and error analysis)
 412 at the same 10-seed standard used throughout the main body. The remaining checks below (A.11
 413 onward) are single-seed diagnostics, not held to that standard; this is noted individually in each.

414 A.1 Related Work

415 **LLM Routing.** LLM routing aims to reduce inference cost by selecting the most appropriate model
 416 for each query. Recent work has shown that routing can significantly improve cost-quality tradeoffs
 417 compared to always using the most capable model. RouteLLM [Ong et al., 2024] learns lightweight
 418 routers from human preference data and dynamically selects between stronger and weaker models at
 419 inference time. ZOOTO [Lu et al., 2024] uses reward-model signals to train routing functions that
 420 dispatch queries to expert models. More recently, Router-R1 [Zhang et al., 2025] formulates routing
 421 as a sequential decision-making problem and trains an LLM router through reinforcement learning,
 422 interleaving reasoning and routing actions. While these approaches learn routing functions directly
 423 from preference labels, reward signals, or reinforcement learning, we instead distill the policy of
 424 an expensive teacher router into a lightweight deployable router — compressing an existing, costly
 425 policy rather than training a new one from preference, reward, or RL signals.

426 **Routing Distillation and Reasoning Supervision.** Knowledge distillation is widely used to transfer
 427 capabilities from large models to smaller ones, for example by using teacher-generated rationales as
 428 auxiliary training supervision [Hsieh et al., 2023]. Reasoning has also been explored as a mechanism
 429 for improving LLM decision making: in routing systems such as Router-R1, reasoning is generated
 430 during inference and contributes directly to routing decisions; closer to our setting, Xu et al. [2025]
 431 train a classifier with an auxiliary reasoning head used only during training and dropped at inference,
 432 improving classifier accuracy without added inference cost. By contrast, we investigate reasoning as a
 433 training-time-only auxiliary supervisory signal specifically for routing-policy distillation, discarded at
 434 deployment, to study whether reasoning improves routing-policy transfer without increasing inference
 435 cost.

436 A.2 Dataset Splits

Table 4. Split sizes and the oracle’s routing-label distribution (percentage of queries whose cheapest-correct tier is small/medium/ large), measured on the training split, independent of any teacher.

Dataset	Train	Validation	Test	Small	Medium	Large
GSM8K	1,000	747	500	6.8%	51.8%	41.4%
MATH	1,000	749	500	4.0%	24.6%	71.4%
Combined	2,000	1,496	1,000	5.4%	38.2%	56.4%

Table 5. Candidate answering models and the teacher router used to generate routing supervision.

Tier	Model	Parameters	Role
Small	Gemma-3-270M	268M	Cheapest candidate; also the distilled models’ shared backbone
Medium	Qwen2.5-0.5B	494M	Medium-cost candidate
Large	Gemma-3-1B	1.0B	Most capable/expensive candidate
Teacher	Qwen2.5-3B-Instruct	3.1B	Prompted only, never fine-tuned; reasoning zero-shot, route few-shot

A.3 Candidate Models

A.4 Teacher Configuration Details

The teacher router follows a two-stage prompting procedure. For each query, a zero-shot prompt first generates a teacher reasoning trace describing the query’s reasoning requirements, and a few-shot prompt then predicts a routing decision conditioned on both the query and the generated reasoning trace. Few-shot demonstrations are sampled from the calibration split (4.3) and remain fixed throughout training and evaluation. The teacher router is Qwen2.5-3B-Instruct.

A.5 Distilled Router Training

We train the two routing-policy distillation variants introduced in Sections 3.5 and 3.6 — route-only (teacher routing labels only) and reasoning-plus-route (additionally, teacher reasoning traces) — with all other training aspects identical, isolating the effect of reasoning supervision. Unless otherwise specified, all distilled routers are trained with the AdamW optimizer [Loshchilov and Hutter, 2019] at a learning rate of 2×10^{-5} , batch size 8, bf16 mixed precision, and 3 training epochs, selected using validation macro-F1 (imbalanced routing labels) and evaluated once on the held-out test split. Each main experiment is repeated across 10 seeds, with results reported as mean $\pm$ standard deviation.

A.6 Evaluation Metric Definitions

Policy Transfer. Policy transfer measures how closely a distilled router reproduces the teacher router’s routing policy. We report accuracy and macro-F1 between distilled routing labels and teacher routing labels. This metric answers RQ1 and RQ2.

Routing Accuracy. Routing accuracy measures agreement with oracle labels rather than the teacher router. Accuracy and macro-F1 are computed against oracle labels and provide an independent measure of routing quality. This metric answers RQ3 by evaluating whether improvements in policy transfer correspond to improvements in routing accuracy.

Deployment Efficiency. We report routing latency and speedup relative to the teacher router. *Routing latency* is the wall-clock time from an already-tokenized query to a routing decision: a single forward pass for the distilled router (3.4), or the sum of the teacher’s two generation calls (A.4). It excludes tokenization, detokenization, model loading, and the candidate’s own *answer-generation* latency, added back only in the cost-quality analysis (Figure 2). Latencies are measured at batch size one on the same GPU (NVIDIA GB10), as the mean over the test split for one run, not seed-averaged like accuracy (A.5).

A.7 Efficiency Across All Configurations

This holds across every evaluated configuration (Tables 1 and 6): label-space granularity has little effect on latency, and although scaling the backbone from 270M to 1B costs more, substantial efficiency gains over the teacher remain.

Table 6. Efficiency, Generalization Checks. Latency and speedup vs. the teacher for the label-space/backbone variants beyond the primary configuration, whose latency and speedup are already reported in Table 1 (5.3, A.8). Teacher latency is measured directly (mean over the pooled 1,000-query test split).

Configuration	Variant	Latency (ms)	Speedup
270M, Three-Label	Route-only	10.6	153×
	Reasoning-plus-route	11.2	145×
1B, Three-Label	Route-only	20.8	78×
	Reasoning-plus-route	22.4	72×
1B, Binary	Route-only	20.8	78×
	Reasoning-plus-route	22.3	73×

A.8 Scaling Analysis

We next examine whether routing policy changes when scaling the distilled-router backbone from 270M to 1B, at both label spaces, completing the full 2×2 grid (label space × backbone). For the most part, it does not.

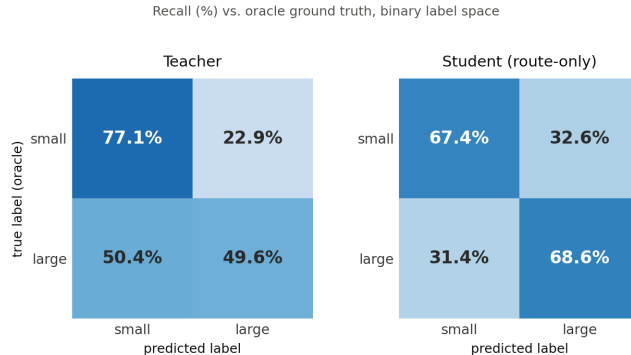
Increasing model size produces only modest, noise-level changes in oracle-measured accuracy in either direction (Oracle F1 column, Table 3): at three-label, route-only and reasoning-plus-route distillation essentially swap rank between 270M and 1B, landing well within one standard deviation of each other throughout; at binary, both variants drift slightly lower from 270M to 1B, a small dip rather than a genuine crossover. In neither label space does increasing router size produce a reliable routing-accuracy advantage.

Interestingly, model size matters for one thing: the association between reasoning supervision and policy transfer grows more apparent at 1B (Table 2), suggesting larger routers better absorb extra supervisory signal even when that signal does not improve routing correctness. Taken together, these results imply routing may be a low-capacity prediction problem for which lightweight models already suffice, with performance limited more by the quality of supervision (5.3) than by model size — encouraging news for deployment.

A.9 Error Analysis

To understand routing policy better, we analyze confusion matrices pooling every test prediction across all 10 seeds and both datasets ($n = 10,000$ per variant) under the binary-routing formulation, against oracle labels. The pattern is consistent: distillation’s gains concentrate in one specific failure mode rather than spreading evenly across all decisions.

Figure 4. Teacher vs. Distilled Router, Both Against Oracle Ground Truth (binary label space). Row-normalized recall; “Distilled Router” is route-only, this paper’s primary configuration. On large-tier queries (bottom row), the teacher is close to a coin flip (49.6% correct), while the distilled router correctly escalates 68.6% of the time.



Against the oracle, the teacher itself under-escalates on large-tier queries at essentially chance level (Figure 4). Both distilled variants correct this specifically, recovering the large majority of those large-tier queries at the cost of trading back a little small-tier recall — this correction explains much of the accuracy improvement reported in Section 5.1. Apart from it, the two variants show highly similar error patterns. A smaller, related asymmetry appears against the teacher’s own decisions rather than the oracle: a distilled model over-escalates roughly twice as often as it under-escalates (42–44% vs. 22–24%), an “err toward the safer, more expensive tier” bias that is separate from, and smaller than, the oracle-referenced correction above. The remaining errors largely reflect ambiguity near routing boundaries, where queries could reasonably go either way.

These findings suggest distillation’s primary benefit is not broad improvement across all routing decisions, but correction of one specific, important failure mode in the teacher.

A.10 Correlation Test Details

We run three successively stricter correlation tests between teacher-agreement F1 and oracle F1 to confirm the null relationship reported in Section 5.5 is not an artifact of the eight-point scatter. A naive Pearson correlation across all eight points is strongly positive ($r = +0.97$, $p < 0.001$), but this turns out to be a structural confound rather than evidence that fidelity predicts correctness: binary-label-space configurations score higher than three-label ones on *both* axes simply because binary is the easier problem (Table 3), inflating the pooled correlation without reflecting any within-configuration relationship. Controlling for label space removes it: a partial Pearson correlation over all 80 individual seed runs (10 seeds $\times$ 8 configuration/variant pairs) residualized against a binary-vs.-three-label indicator collapses to $r = -0.16$ ($p = 0.17$, $n = 80$). A more direct test confirms this, correlating the reasoning-vs.-route-only *change* in fidelity against the corresponding change in oracle F1, one point per configuration ($n = 4$): Pearson $r = -0.45$ ($p = 0.55$); Spearman $\rho = 0.00$ ($p = 1.0$) — which configurations gain the most fidelity carries no information about which gain the most accuracy. All three tests agree once the label-space confound is removed.

A.11 Loss Weight Sweep

$\lambda_{\text{reason}} \in \{0.5, 1.0\}$ was swept at the 1B backbone, three-label label space, single seed (42), best epoch per setting (validation macro-F1): $\lambda_{\text{reason}} = 0.5$ reaches 0.548, $\lambda_{\text{reason}} = 1.0$ reaches 0.541. 0.5 was selected and used throughout this paper’s reasoning-plus-route runs.

A.12 Seed Sensitivity

An initialization bug (the classification head was built before the training seed was applied) produced a 0.060 macro-F1 spread between two nominally identical runs, roughly ten times the route-only/reasoning-plus-route gap at the primary configuration (Table 2). Fixed by seeding before model construction; even after the fix, two seeds at the same setting still differed by 0.023: the motivation for the 10-seed protocol used throughout this paper.

A.13 Epoch-Count Robustness

Every main-body result uses a 3-epoch budget (A.5). We checked this at a single seed (42), 20 epochs, both label spaces.

Both label spaces peak well past epoch 3 (binary: route-only at epoch 2, reasoning-plus-route at epoch 6; three-label: epoch 5 and 8 respectively), with validation macro-F1 gains of 0.08–0.12 over the 3-epoch checkpoint. On held-out test data, though, the gain is smaller and inconsistent: route-only/binary (this paper’s primary configuration) is slightly *worse* at its best 20-epoch checkpoint (0.640) than the main protocol’s 10-seed mean (0.651 ± 0.005); the other three configurations indicate modest gains within roughly one standard deviation.

Because route-only/binary is this paper’s primary configuration, we repeated it at 20 epochs across the same 10 seeds used throughout the main protocol, rather than relying on the single-seed check above. The 20-epoch test macro-F1 is 0.653 ± 0.013 , against the 3-epoch main-protocol result of 0.651 ± 0.005 (Table 2). A paired seed-by-seed bootstrap over the 10 matched seeds gives a mean improvement of $+0.002$ macro-F1 with a 95% CI of $[-0.006, +0.011]$, which includes zero: six

541 of the 10 seeds are lower at 20 epochs than at 3 epochs, and seed-to-seed variance nearly triples
542 (± 0.013 vs. ± 0.005). Extending training therefore does not yield a reliable test-set improvement at
543 this configuration, despite the consistent validation-macro-F1 gains observed above; this confirms the
544 single-seed result was not an artifact of that one seed, and supports retaining the 3-epoch, 10-seed
545 protocol as this paper’s main result.

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Justification: Table 2 reports mean $\pm$ SD across ten independently seeded training runs per variant and configuration (Section A.5), and the same ten seeds are used for a paired bootstrap [Koehn, 2004] on the route-only-vs.-reasoning-plus-route macro-F1 difference, reported with a 95% CI and an explicit statement of whether that CI excludes zero (Table 2). This identical protocol is applied across all four configurations (two label spaces $\times$ two backbone scales, Section A.8), not just the primary one. Appendix A.12 additionally reports the parameter-initialization seed spread (0.023 macro-F1 between nominally identical runs) that motivated the 10-seed protocol, and Appendix A.13 reports a further paired seed-by-seed bootstrap when checking robustness to training-epoch count. All reported error bars are sample standard deviations across the 10 seeds, not standard errors.

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Justification: Section A.5 states both variants trained on a single NVIDIA GB10 GPU for 3 epochs at batch size 8 in bf16; Section A.6 and Table 1 report per-query inference latency measured on the same GPU. We do not report a total-project compute estimate including preliminary or failed runs.

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